Grid L

time limit per test

2 seconds

memory limit per test

256 megabytes

Roger has p unit-length segments and q L-shaped pieces, each formed by joining two unit-length segments at a right angle.

Unit-length segment

L-shaped piece
(two unit-length segments at 90°)

He wants to use all of these pieces, with no piece left unused, to form a grid of dimensions $n \times m$.

Given p and q, determine whether there exist positive integers n and m such that an $n \times m$ grid can be formed using exactly p unit-length segments and q L-shaped pieces, possibly rotating them, and provide any valid n and m.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The first line of each test case contains two integers p and q ($1 \le p,q \le 108$).

Output

For each test case, print any valid n and m such that it's possible to construct $n \times m$ using all pieces. If there are no such n and m, print -1.

Example

Input

7

1 2

1 3

5 1

2 5

2 10

100000000 100000000

1 1

Output

-1

1 2

1 2

2 2

2 4

-1

-1

Note

Here are the constructions for examples 2, 3, and 4. We use a different color for different L-shaped pieces, black for all unit-length segments.